

RYGGE MOSS, Norway

WMO: 014940

Lat: 59.379N

Lon: 10.786E

Elev: 53

StdP: 100.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.6	-13.0	-17.6	0.8	-14.3	-15.1	1.0	-11.2	11.4	4.8	10.4	4.6	1.4	0	0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	25.9	17.4	24.0	16.7	22.2	16.1	19.1	23.1	18.1	22.0	17.3	20.6	3.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	12.8	20.3	16.8	12.1	19.4	15.9	11.4	18.7	54.2	23.0	51.4	21.9	48.7	20.6	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.4	7.4	-18.2	28.2	3.6	2.4	-20.8	29.9	-22.9	31.3	-25.0	32.6	-27.6	34.3		
(5)			-18.5	20.3	3.5	0.8	-21.0	20.9	-23.1	21.4	-25.1	21.9	-27.6	22.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-2.3	-1.8	1.2	5.9	10.8	14.6	17.4	16.4	12.5	7.3	2.7	-0.9		
	DBStd	7.89	5.15	4.96	3.84	3.15	3.42	2.52	2.43	2.39	2.77	3.47	4.05	5.43		
	HDD10.0	1809	382	330	273	129	30	1	0	0	10	97	221	338		
	HDD18.3	4162	640	563	531	373	236	116	48	71	175	342	470	596		
	CDD10.0	722	0	0	0	6	54	138	228	198	85	13	1	0		
	CDD18.3	33	0	0	0	0	1	4	18	10	0	0	0	0		
	CDH23.3	226	0	0	0	0	13	36	131	46	0	0	0	0		
	CDH26.7	30	0	0	0	0	1	3	24	2	0	0	0	0		
(14) Wind	WSAvg	3.4	3.3	3.4	3.4	3.5	3.6	3.6	3.3	3.2	3.4	3.4	3.4	3.2		
(15) Precipitation	PrecAvg	913	78	58	45	56	58	74	79	100	84	106	98	78		
	PrecMax	1269	176	145	139	116	109	145	158	185	213	272	305	160		
	PrecMin	694	13	16	0	11	4	8	5	40	27	16	26	19		
	PrecStd	151	41	33	30	27	27	31	34	38	43	59	60	43		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	8.6	13.8	19.2	25.3	26.9	29.1	26.8	21.3	15.5	12.0	9.4		
		MCWB	6.0	4.8	7.7	11.4	16.0	16.8	18.9	17.6	15.1	12.2	10.2	8.4		
	2%	DB	6.3	6.3	10.2	16.1	21.9	24.1	26.8	24.7	19.2	14.1	10.8	8.2		
		MCWB	5.0	4.2	5.7	9.7	13.7	15.4	18.1	17.7	14.4	11.8	9.6	7.3		
	5%	DB	5.2	5.2	7.9	13.6	19.2	21.8	24.4	22.9	18.0	13.2	9.6	7.2		
		MCWB	4.2	3.8	4.0	8.2	12.6	14.6	17.2	17.1	14.1	11.5	8.5	6.2		
	10%	DB	4.2	4.4	6.1	11.4	17.0	19.8	22.5	21.0	16.9	12.3	8.4	6.1		
		MCWB	3.3	3.2	3.5	6.9	11.3	13.7	16.7	16.3	13.7	10.8	7.4	5.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	6.1	8.2	12.8	17.1	18.4	20.4	20.1	17.4	13.7	11.2	8.9		
		MCDB	7.2	6.9	12.9	17.5	22.0	23.1	25.6	23.0	19.0	14.4	11.5	9.3		
	2%	WB	5.5	5.1	6.1	10.2	15.1	16.9	19.3	18.9	16.2	12.9	9.8	7.7		
		MCDB	5.9	5.6	8.8	14.6	19.5	21.9	24.3	22.1	17.7	13.5	10.3	8.1		
	5%	WB	4.4	4.3	5.1	9.0	13.6	15.8	18.5	18.0	15.2	12.1	8.7	6.4		
		MCDB	4.9	4.9	6.7	12.3	17.6	19.6	22.8	21.4	16.9	12.8	9.3	6.9		
	10%	WB	3.4	3.4	4.1	7.8	12.4	14.8	17.6	17.1	14.3	11.2	7.6	5.2		
		MCDB	4.1	4.1	5.6	10.4	16.1	18.3	21.2	20.0	16.1	12.2	8.3	5.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	5.5	6.9	7.9	8.3	7.7	7.7	7.5	7.1	5.8	4.6	4.8		
		MCDBR	5.0	5.4	9.4	11.7	12.2	11.5	11.5	10.0	8.4	5.9	5.4	4.4		
	5% WB	MCWBR	4.7	4.3	5.9	6.5	5.9	5.2	5.0	4.8	5.1	4.6	5.4	4.5		
		MCWBR	4.3	4.7	7.9	9.8	10.0	9.4	8.4	8.4	6.5	4.9	5.0	4.4		
(40) Clear-Sky Solar Irradiance	taub		0.278	0.314	0.335	0.374	0.369	0.357	0.378	0.373	0.343	0.331	0.295	0.253		
	taud		2.234	2.317	2.366	2.322	2.411	2.463	2.425	2.434	2.506	2.440	2.294	2.154		
	Ebn at Noon		526	699	802	827	861	877	846	822	793	677	500	409		
	Edh at Noon		43	67	86	106	104	101	102	94	75	60	40	31		
(44) All-Sky Solar Radiation	RadAvg		0.30	0.89	2.40	3.80	5.05	5.78	5.16	4.07	2.71	1.20	0.41	0.21		
	RadStd		0.06	0.16	0.26	0.53	0.50	0.39	0.63	0.29	0.30	0.22	0.10	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page